

Equivalent Fractions and Comparing — Look at Both Numbers

A fraction has two jobs to read. The **bottom** number says how many equal pieces the whole was cut into, so a bigger bottom number means *smaller* pieces. The **top** number says how many of those pieces you have.

- Looking at only the tops, or only the bottoms, is not a way to compare two fractions. It is right only when the other number already matches.
- When neither number matches, rewrite both fractions with the same bottom number, then compare the tops.
- Draw a picture or a number line whenever it helps. Show your work.

1. The bottom numbers already match, so the pieces are the same size. Write $<$, $>$, or $=$ in the blank:

$$\frac{3}{8} \quad \text{—————} \quad \frac{5}{8}$$

Then write one sentence saying why counting the top numbers is safe here.

2. This time the *top* numbers match, so you are holding the same number of pieces — but the pieces are not the same size. Write $<$, $>$, or $=$:

$$\frac{3}{5} \quad \text{—————} \quad \frac{3}{8}$$

3. Find and fix. Ben writes $\frac{2}{9} > \frac{2}{4}$, and says, “Nine is bigger than four.”

Ben compared only the bottom numbers. Fix his answer: write $<$, $>$, or $=$ in the blank, and say in a few words what a bigger bottom number really does to the pieces.

$$\frac{2}{9} \quad \text{—————} \quad \frac{2}{4}$$

4. Find and fix. Cara wants to rewrite $\frac{3}{5}$ with 40 on the bottom. She writes $\frac{3}{40}$, changing only the bottom number.

Whatever you do to the bottom, you must do to the top. Work out the correct top number for $\frac{\square}{40}$ and write it on the answer line.

Answer: _____

5. Which is greater, $\frac{3}{4}$ or $\frac{5}{7}$? Neither number matches, so counting tops or bottoms cannot decide it.

Rewrite both fractions with 28 on the bottom. Write the top number that $\frac{3}{4}$ becomes on the answer line, and write $<$, $>$, or $=$ in this blank:

$$\frac{3}{4} \quad \text{—————} \quad \frac{5}{7}$$

6. Write $\frac{6}{8}$ in its simplest form — the equal fraction with the smallest possible numbers. Divide the top and the bottom by the same number.

Answer: _____

7. Put these three fractions in order from **smallest to largest**:

$$\frac{2}{3}, \quad \frac{3}{5}, \quad \frac{7}{10}$$

Rewriting all three with 30 on the bottom will settle it. Write the three fractions in order on the answer line.

Answer: _____

8. Explain, then compare.

(a) A friend says, “To compare two fractions, just look at the top numbers.” Explain, in your own words, why that does not work, and give one example where the friend’s shortcut gives the wrong answer.

(b) Compare $\frac{5}{6}$ and $\frac{7}{9}$ by rewriting both with 18 on the bottom. Write the top number that $\frac{5}{6}$ becomes on the answer line, and write $<$, $>$, or $=$ in this blank:

$$\frac{5}{6} \quad \text{_____} \quad \frac{7}{9}$$